

Aditya ML, DL, NLP
course

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(1)

Soc 32

K-Nearest
Neighbour
ML Algorithm

Lec 1 :- KNN In-depth
intuition

Lec 2 :- Optimization
of KNN :-

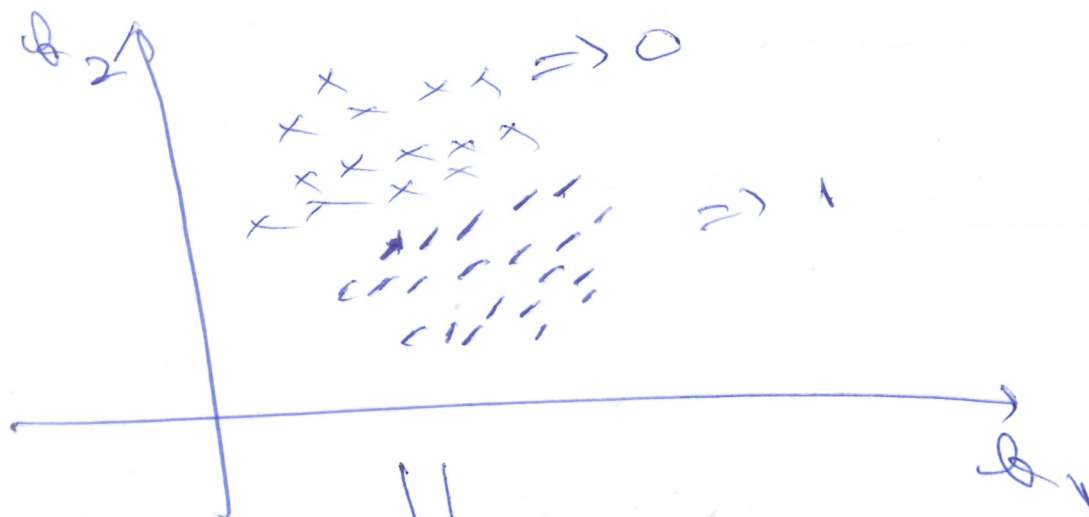
KD tree and Ball
tree In-depth.

K-Nearest Neighbour

- (1) Classification
- (2) Regression

① Classification :-

②



Training

① We have to initialize the K -value

$K > 0$ ----- $\square K=5$
 $K = 1, 2, 3, 4, 5, \dots$
 $\Downarrow$

Hyper-parameter tuning.

② Find the value of K Nearest Neighbour for the test.

3 3rd Step:-

(3)

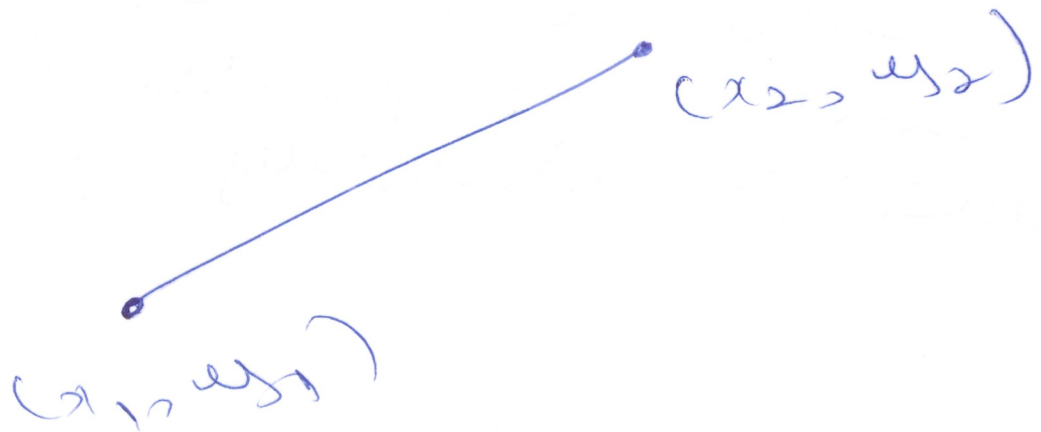
From those $K=5$, now
many neighbors belong
to 0 category and 1
category.

0's $\rightarrow 2$

1's $\rightarrow 3$

$\hookrightarrow$ Maximum Neighbors
belong to 1's category
hence, here the output
is 1.

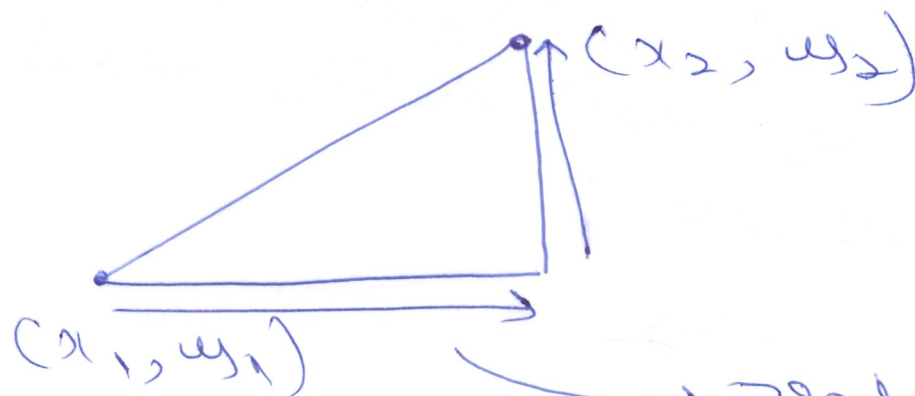
$\Rightarrow$ Euclidean Distance



$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

② Manhattan Distance

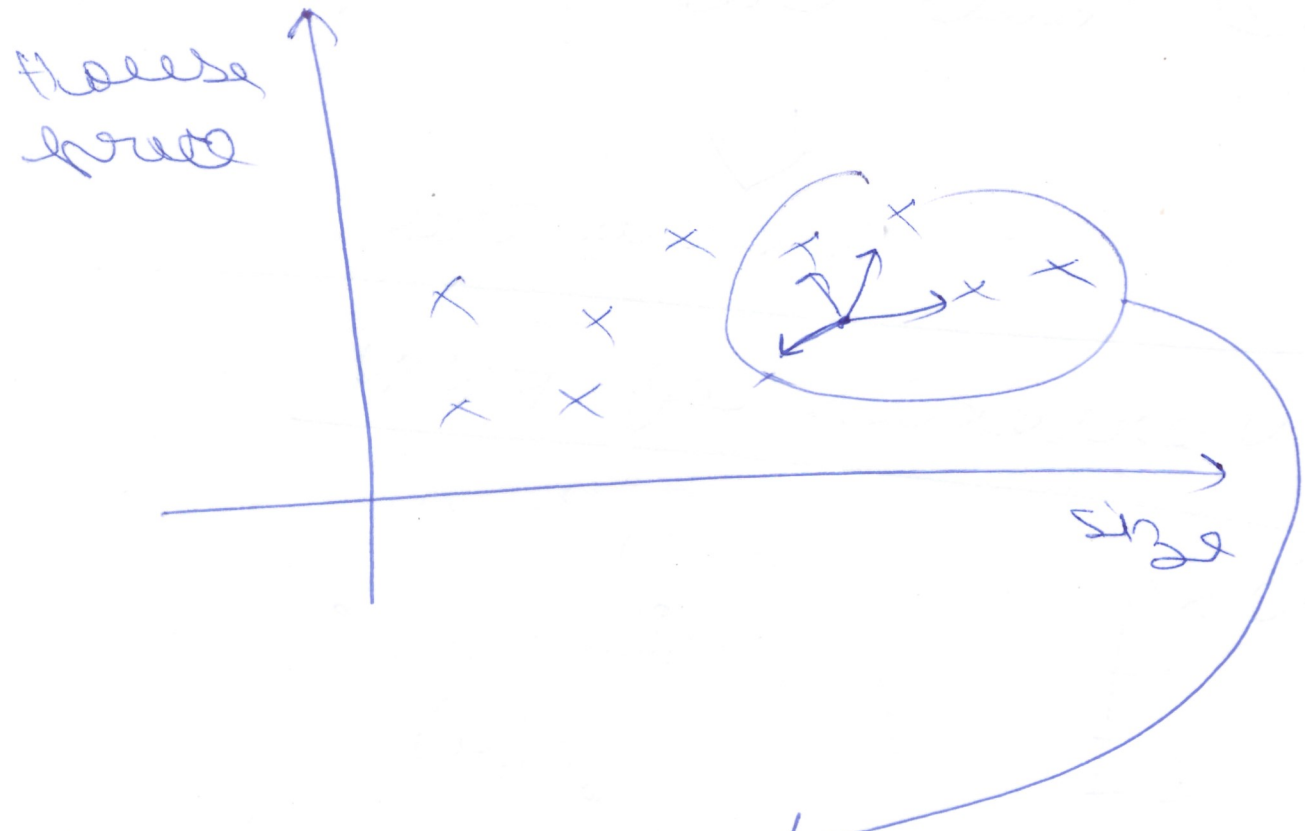
④



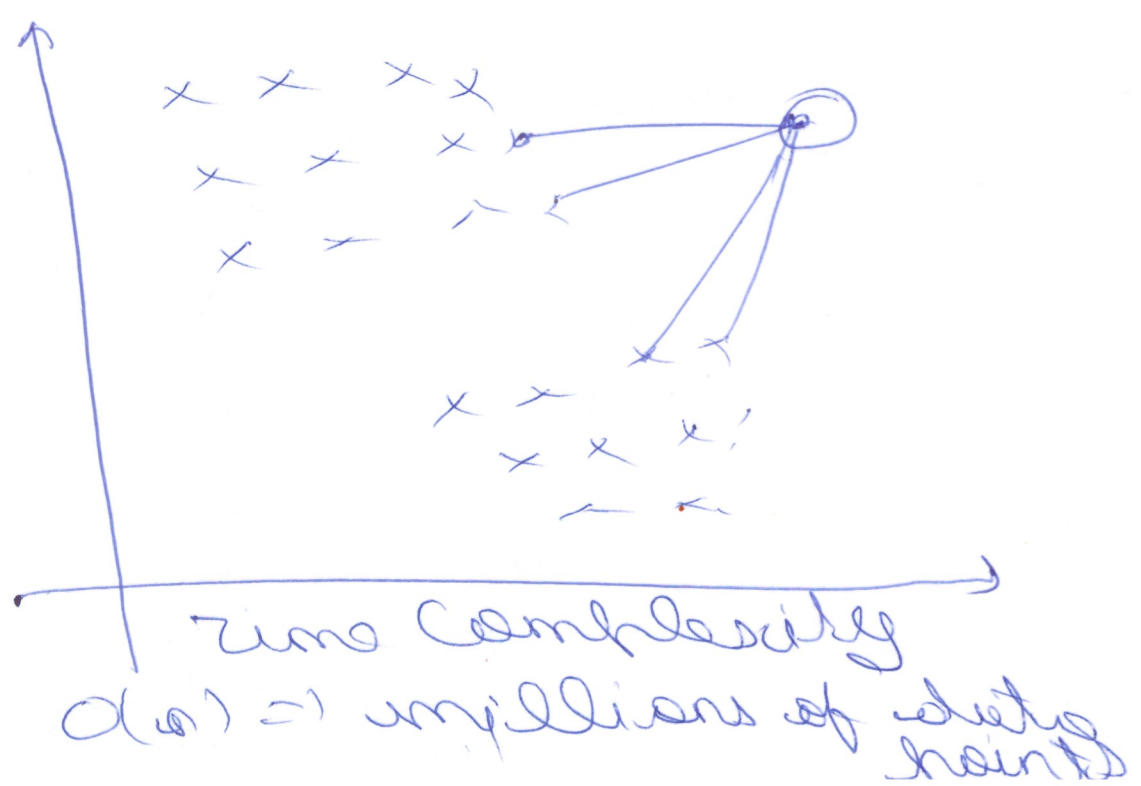
→ This total distance is basically the Manhattan distance

→ Here first we have find distance between them & tell at length that which belongs to which category.

=> Regression



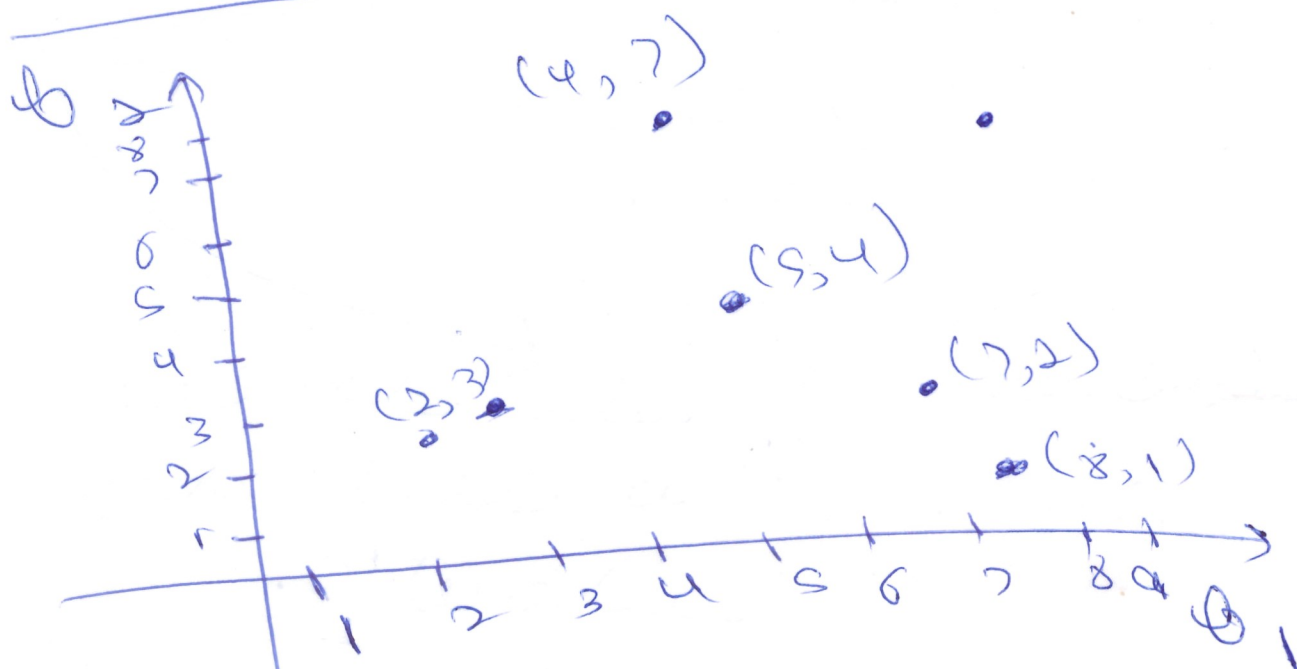
Average of all these points to find the output



- (1) B+ tree } $\Rightarrow$ optimal
(2) B-tree } $\Rightarrow$ optimal

$\Downarrow$
Binary tree

$\Rightarrow$ Variants of FNN :-



b₁

?

5

4

2

4

8

median

b₂

2

4

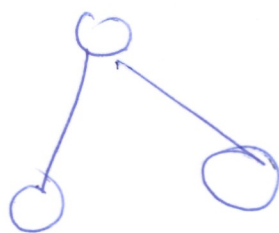
6

3

7

1

median



7

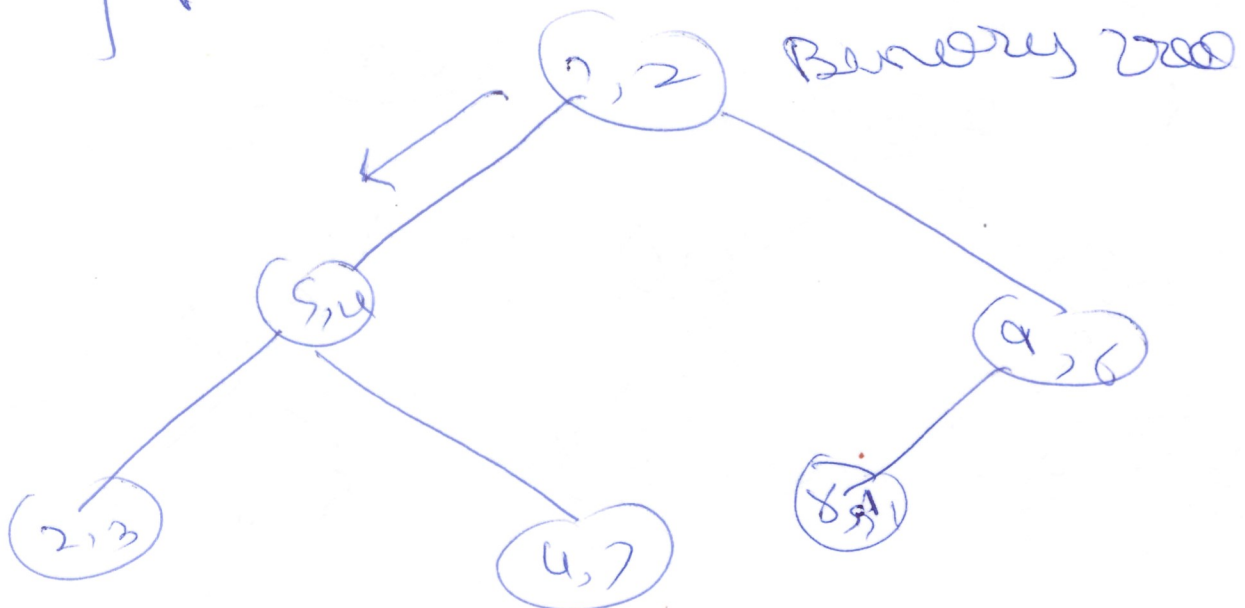
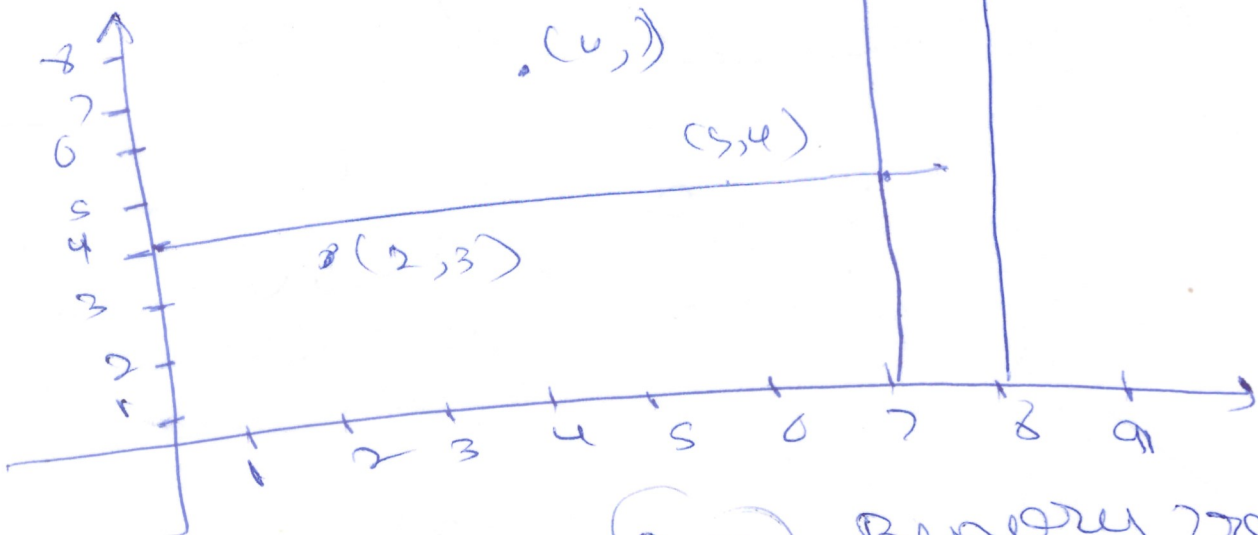
$$\frac{81}{2, 4, \boxed{5, 7}, 8, 9}$$

$$\frac{5+7}{2} = \frac{12}{2} = 6$$

Q2 :-

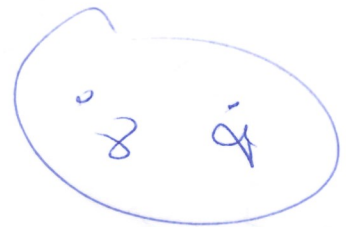
$$2, \boxed{3, 4}, 6, 7$$

$$\Rightarrow \text{node} = \frac{3+4}{2} = 3.5$$



2 Bell tree

8



1 2 3 4

we start
overalmenly
from here

